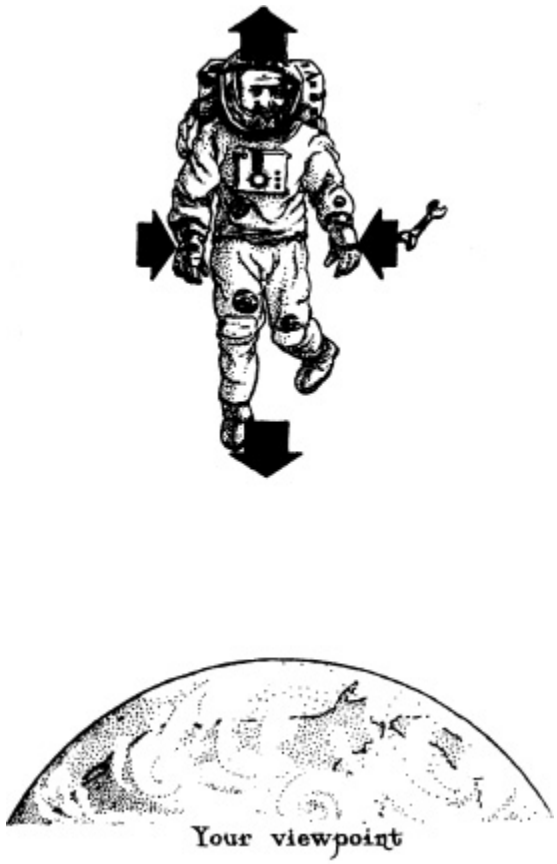


2.3 As you fall toward Earth, tidal gravitational forces stretch you from head to foot and squeeze you from the sides.

As seen from Earth (Figure 2.3a), the gravitational pull is slightly different on various parts of your body. Because your feet are closer to the Earth than your head, gravity pulls more strongly on them than on your head, so it stretches you from foot to head. And because gravity pulls always toward the Earth's center, a direction that is slightly leftward on your right side and slightly rightward on your left side, the pull is slightly leftward on your right and slightly rightward on your left; that is, it squeezes your sides inward.

From your viewpoint (Figure 2.3b), the large, downward force of gravity is gone, vanished. You feel weightless. However, the vanished piece of gravity is only the piece that pulled you downward. The head-to-foot stretch and side-to-side squeeze remain. They are caused by the *differences* between gravity on the outer parts of your body and gravity at your body's center, differences that you cannot get rid of by falling freely.

The vertical stretch and lateral squeeze that you feel, as you fall, are called tidal gravity or tidal gravitational forces, because, when the Moon is